

Name: Balaji Udayakumar

Email: Balajiudayakumar@outlook.com

Phone: (603) 584-1415

LinkedIn: <https://www.linkedin.com/in/u-balaji-a2263728b/>

INTRODUCTION

Experienced Python Developer with a robust background in software development, data analysis, and system integration. Adept at working in dynamic environments, transforming complex requirements into efficient, scalable solutions. Proven ability to deliver projects within deadlines while maintaining high standards of quality.

PROFESSIONAL SUMMARY

Results-oriented Software Developer with 6+ years of experience in designing, developing, and deploying enterprise applications. Specialized in building highly scalable backend services using Python. Strong understanding of system architecture, data structures, algorithms, and database management. Notable experience in **warehouse management** technology and real-time data processing.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, SQL, HTML, CSS
- **Web Technologies:** React.js, Django, Flask, RESTful APIs, SOAP
- **Database Management:** Oracle SQL, PostgreSQL, MySQL, MongoDB
- **Cloud Services:** AWS (EC2, S3, Lambda, RDS, IoT), Google Cloud Platform
- **DevOps Tools:** Docker, Kubernetes, Jenkins, GitLab CI/CD
- **Project Management Tools:** Jira, Trello, Azure DevOps
- **Version Control:** Git, GitHub, Bitbucket
- **Agile Methodologies:** Scrum, Kanban

PROFESSIONAL EXPERIENCE

Software Development Engineer

AMAZON ROBOTICS – Reading, MA | June 2023 – April 2024

- Developed and maintained scalable web applications using **Django** and **Flask**, enhancing user experience and system performance.

- Led the design and architecture of systems with a focus on design patterns, reliability, and scalability.
- collaborated with **cross-functional teams** to design and architect scalable systems, ensuring alignment with business goals and user needs.
- Implemented **machine learning algorithms** for data analysis projects, improving predictive accuracy and operational efficiency.
- Conducted code reviews and knowledge-sharing sessions at Amazon Robotics to enhance **team** capabilities and maintain high standards of code quality
- Designed RESTful APIs, ensuring seamless data integration across various services.
- Optimized database performance and ensured data integrity using **Oracle SQL** and **PostgreSQL**.
- Migrated legacy systems to cloud-based solutions on **AWS**, resulting in improved scalability and reduced costs.
- Utilized **Docker** and **Kubernetes** for containerization and orchestration, streamlining the deployment process.

Software Development Engineer (Internship)

AMAZON ROBOTICS – Reading, MA | October 2022 - May 2023

- Implemented machine learning algorithms for data analysis projects, improving predictive accuracy and operational efficiency.
- Experienced in all phases of the software development life cycle, from requirement analysis to deployment and maintenance.
- Conducted thorough code reviews to ensure adherence to best practices and to improve overall code quality and **team** knowledge sharing.
- Managed and optimized software operations, including monitoring, troubleshooting, and maintaining production systems for maximum uptime and performance.

Software Developer

Tata Consultancy Services (TCS) - Chennai, Tamil Nadu, India | April 2017 – December 2021

- Conducted data analysis using **Python**, translating raw data into actionable insights for decision-making.

- Managed projects in web development by adopting **Agile methodologies**, optimizing **team** performance and project deliveries.
- Applied advanced statistical models to uncover data patterns and forecast trends, driving strategic initiatives.
- Engaged in pair programming sessions to tackle complex coding challenges and foster a culture of peer learning and support within the **team**.
- Managed databases, ensuring data integrity through **SQL** querying and **ETL** processes.
- Conducted quality assurance activities to identify and resolve issues, ensuring deliverables met quality standards.

EDUCATION

- Master in computer information systems, Rivier University-Nashua, New Hampshire, United States of America.
- Bachelor of Engineering in Computer Science and Engineering, Anna University (Meenakshi College of Engineering), Chennai, Tamil Nadu, India.

PROJECTS

E-commerce Web Application

- **Description:** Developed a full-stack e-commerce web application using **Django**, **React.js**, and **PostgreSQL**.
- **Responsibilities:** Designed the backend architecture, implemented **RESTful APIs**, integrated payment gateways, and optimized database queries for faster load times.

Real-Time Data Analytics Platform

- **Description:** Built a real-time data analytics platform using **Flask**, **Apache Kafka**, and **Pandas** for streaming data processing.
- **Responsibilities:** Developed data ingestion pipelines, implemented machine learning models for predictive analytics, and created interactive dashboards for **data visualization**.

Banking Fraud Detection System

- **Description:** Developed a banking fraud detection system using **Python**, **AWS**, and **machine learning algorithms**.

- **Responsibilities:** Designed and implemented fraud detection algorithms, developed data processing pipelines on **AWS**, and created a web interface for real-time monitoring and alerting.

Autonomous Warehouse Robotics System

- **Description:** Developed an autonomous robotics system for warehouse management using **AWS RoboMaker** and **ROS (Robot Operating System)**.
- **Responsibilities:** Programmed robotic movement and navigation algorithms, integrated AWS services for real-time data processing and control, and conducted performance testing to ensure reliability and efficiency in a live warehouse environment.

ADDITIONAL INFORMATION

- **Languages:** English (fluent), Tamil (native)
- **Hobbies:** Open-source contributions, tech blogging, participating in hackathons